Smart India Hackathon (SIH 2026) - Project Proposal & Implementation Guide

1. Metadata & Project Overview

Field	Details
Problem Statement ID (PS ID)	SIH26011
Project Title	3D ULPIN Generation and Vertical Property Mapping System
Sponsoring Organization	Ministry of Rural Development / Department of Land Resources (DoLR), Government of India
Category	Software
Primary Target Users	Revenue Departments, Municipal Corporations, Town Planning Authorities, Lending Institutions, Homebuyers, and Developers

2. Problem Definition & Real-World Ground Challenges

Under the Digital India Land Records Modernization Programme (DILRMP), India implemented the Unique Land Parcel Identification Number (ULPIN), also known as Bhu-Aadhaar. This 14-digit alphanumeric code uniquely identifies land based on geographic coordinates (latitude and longitude). However, this existing cadastre is strictly two-dimensional (2D) and surface-bound.

The Critical Bottleneck

- **Vertical Urban Expansion:** With rapid urbanization in major Indian metropolitan regions, single ground-level land parcels now host multi-story residential towers, commercial malls, and high-density apartment complexes. While there is only one surface parcel, it accommodates hundreds of separate property owners vertically.
- **Absence of 3D Volumetric Legal Identifiers:** Existing state revenue records (e.g., Record of Rights, 7/12 extracts, and Bhu-Naksha) cannot uniquely map or delineate properties stacked along the vertical (Z-axis). An apartment unit on the 4th floor has no distinct geographical boundary in current digital land maps.
- **Pervasive Real Estate Disputes & Financial Fraud:**
    - **Duplicate Mortgage Fraud:** Unscrupulous sellers take loans from multiple banks on the same flat using forged paper titles, as banks lack a centralized 3D cadastral registry to verify prior liens.
    - **Common Area & Parking Disputes:** Unregulated sale of stilt parking, basement slots, and terrace access rights leads to ongoing civil litigation.
    - **Subsurface & Air Rights Ambiguity:** Metro tunnels, underground utility networks, and elevated transit corridors intersect private land rights without clear 3D boundary demarcation.

3. Proposed Solution Architecture

The proposed system upgrades India's 2D land record framework into an interoperable, cloud-native **3D Vertical Land Administration Domain (3D-VLAD)**.

Core Architectural Modules

1. **Algorithmic 3D ULPIN Generator:** Extends the statutory 14-digit surface ULPIN with volumetric parameters conforming to the international standard ISO 19152 (Land Administration Domain Model - LADM).
2. **Interactive 3D WebGIS Cadastre Viewer:** A browser-based, client-side 3D model visualizer powered by Three.js that renders multi-story structures, basement infrastructure, and distinct floor-level apartment units.

3. **Spatial Overlap & Conflict Detection Engine:** Automated geometric validation that prevents overlapping boundaries, conflicting allocations of parking slots, or duplicate ownership claims in 3D coordinate space.
4. **Administrative & Citizen Portal:** Role-based dashboard enabling citizens to inspect property titles in 3D and providing revenue officers with tools to approve mutations, subdivisions, and registrations.

4. 3D ULPIN Formulation & Numbering Specification

The 3D ULPIN formula systematically encodes geospatial location, elevation, and unit identification:

`3D-ULPIN = [Base 14-digit 2D ULPIN] + [Floor Level] + [Unit Designation] + [Z-Elevation Ceiling & Floor in Meters]`**Example breakdown:**
`14MH2704291845-FL04-U402-Z12.5-15.5`

- `14MH2704291845`: Base statutory surface land parcel identifier.
- `FL04`: Vertical floor level (Floor 4).
- `U402`: Registered apartment unit number (Unit 402).
- `Z12.5-15.5`: Elevation bounding envelope (from 12.5m to 15.5m above mean ground level).

5. 100% Free Open-Source Student Tech Stack (Zero Budget)

Designed specifically for student developers with no external cloud or licensing expenditure:

- **3D Graphics & Rendering:** Three.js / React Three Fiber (client-side WebGL rendering; requires zero expensive GPU cloud servers).
- **Geospatial & Building Footprints:** OpenStreetMap / Overpass API (free open-access 3D building outlines and polygonal GeoJSON data).

- **Database & Spatial Backend:** Supabase free tier (PostgreSQL with PostGIS extension for spatial queries and real-time subscription listeners).
- **Frontend Web Framework:** Next.js / React with Tailwind CSS.
- **Hosting & Deployment:** Vercel (unlimited free hosting for web apps and serverless API endpoints).

6. 36-Hour Hackathon Execution Roadmap

Timeframe	Key Milestones & Activities
Hours 0–6	**Data Modeling & Schema:** Set up the PostgreSQL schema on Supabase; extract 3D building footprints via Overpass API into GeoJSON; define the 3D ULPIN string generation logic.
Hours 6–18	**3D Visualizer Core:** Implement the Three.js canvas; load floor-by-floor parametric bounding boxes; enable mouse raycasting for individual unit selection on click.
Hours 18–24	**Database Integration:** Connect 3D unit meshes to Supabase; fetch live owner details, mortgage status, and carpet area upon unit selection.
Hours 24–30	**Verification & Conflict Engine:** Build the overlap detection logic to flag duplicate parking allocations and unauthorized terrace privatizations.
Hours 30–36	**Polish & Demo Rehearsal:** Optimize frame rates, test on mobile/tablet viewports, deploy to Vercel, and

Timeframe	Key Milestones & Activities
	rehearse the 30-second live demonstration.

7. The 30-Second Winning Pitch & Live Demo Script

1. **Step 1 (The Hook):** Open the live web application showing a city parcel on OpenStreetMap. Click the parcel to watch an interactive 3D multi-story apartment building extrude in real-time.
2. **Step 2 (Judge Interactivity):** Ask the evaluator: *"Sir/Ma'am, which floor would you like to inspect?"* When they select Floor 7, click that specific level.
3. **Step 3 (Volumetric Delineation):** Unit 701 highlights in vibrant green. The right-hand panel instantly displays:
    - *3D-ULPIN:* `27-NSK-4291-FL07-U701`
    - *Elevation:* Z = 21.0m to 24.0m
    - *Verified Owner:* Legal record linked with municipal property tax ID
    - *Encumbrance Status:* Clean title (no active mortgage disputes)
4. **Step 4 (The Fraud Prevention Demo):** Click Basement Parking Slot P-12. Show the system rejecting a duplicate registration attempt with a red alert: *"Conflict Detected: Slot P-12 already legally assigned to Unit 302."*

8. Socio-Economic & Governance Impact

- **Banking Sector Transparency:** Provides lending institutions with a single, verifiable 3D collateral map, preventing multi-lender mortgage fraud.
- **Consumer Protection:** Ensures homebuyers receive clear, uncontested title to apartments, parking spots, and undivided shares of common amenities.
- **Municipal Revenue Enhancement:** Automates volumetric property tax assessments, enabling city corporations to accurately detect unassessed vertical expansions.